



Automatic Layer Selection for Hallucination Detection

Xinpeng Wang¹ · William Cao² · Andrew Gordon Wilson³ · Zhe Zeng¹

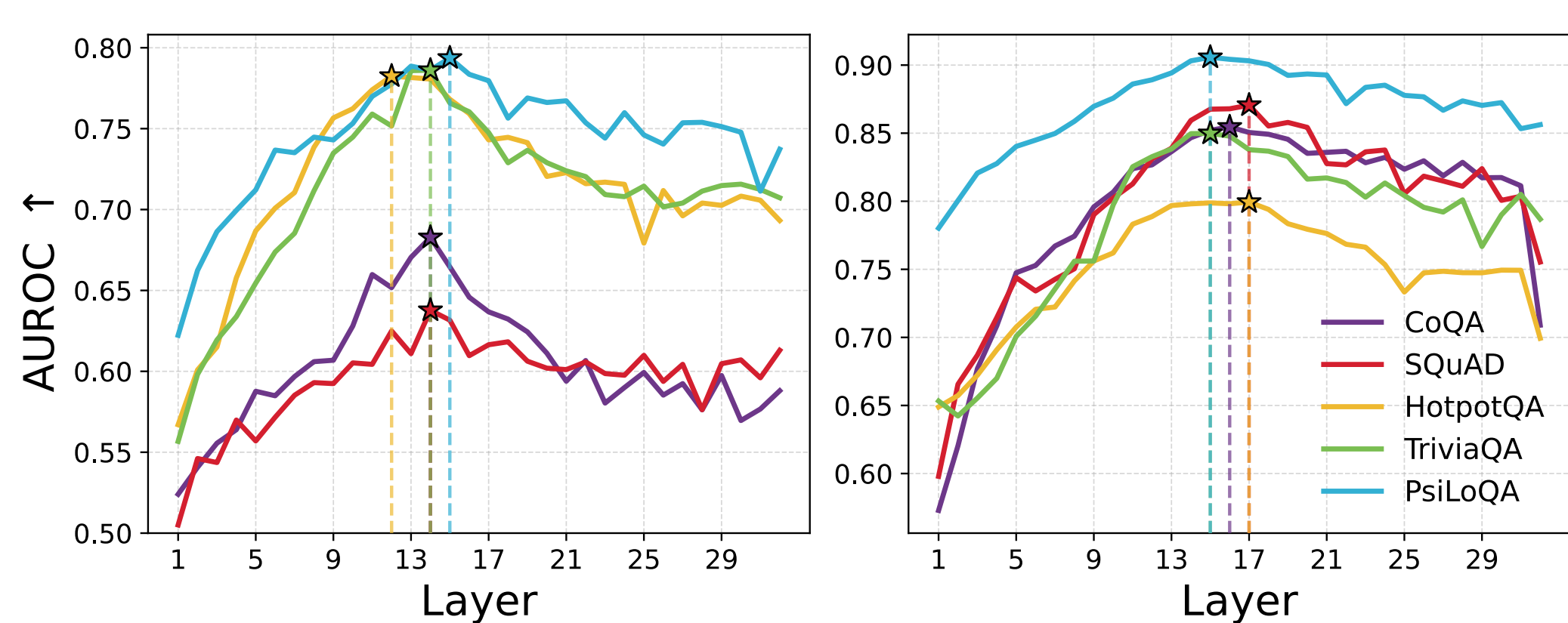
¹University of Virginia · ²Walrus Security · ³New York University

1 First **systematic evaluation** of layer-selection criteria — information-theoretic, gradient-based, and geometric — for hallucination detection.

2 **FEPoID**: a simple, efficient criterion that auto-selects near-optimal intermediate layers across datasets, scales, and architectures.

3 **First-Sentence Truncation (FST)**: probe the end of the first sentence — a method-agnostic boost across every baseline tested.

Q1: Which Layer Should We Extract ?



(a) LLaMA-3.1-8B-Instruct (b) Mistral-7B-Instruct-v0.3
Best-performing layer **varies** across models and datasets

Criteria We Evaluated

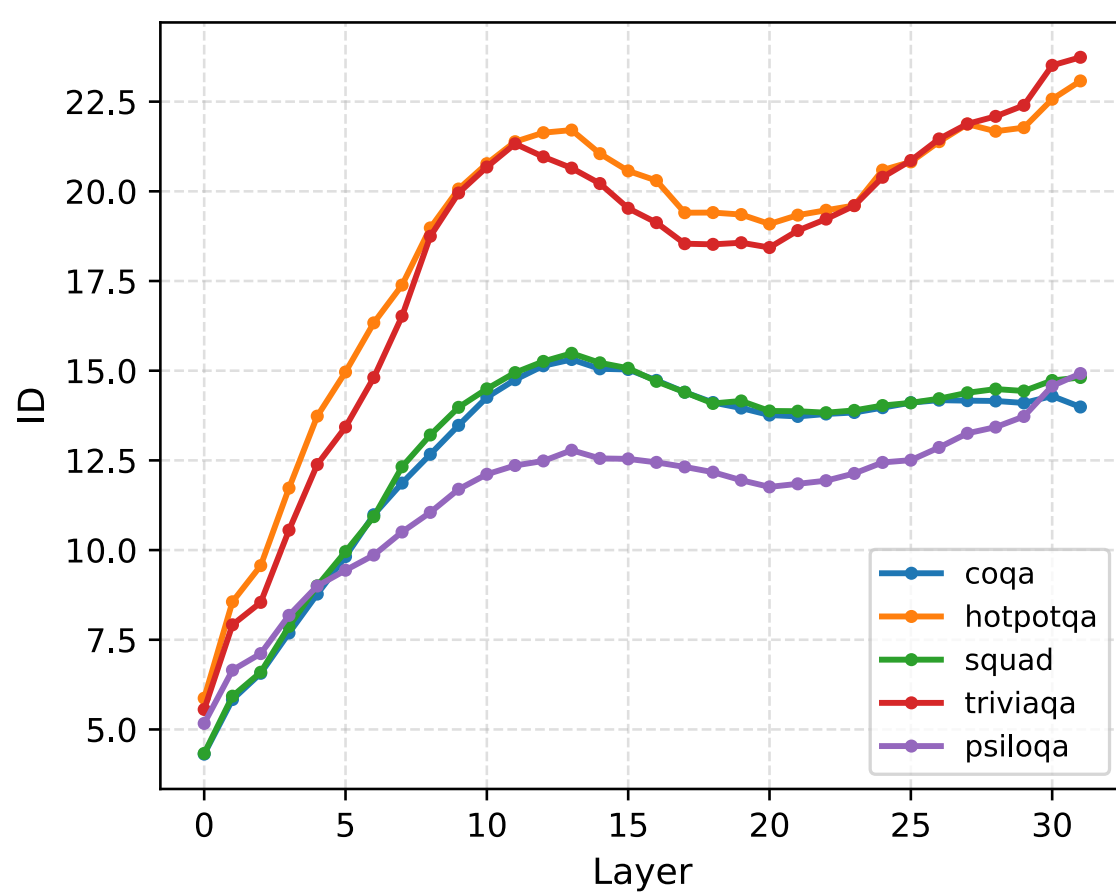
Hypothesis	Criteria
(i) Rich semantic information	RankME
(ii) Task aligned features	Validation Loss
(iii) Information compression	Curvature
(iv) High effective information capacity	Intrinsic Dimension (ID); FEPoID

None of existing criteria can efficiently and reliably pick the best layer

Criteria We Proposed

First Effective Peak of Intrinsic Dimension (FEPoID)

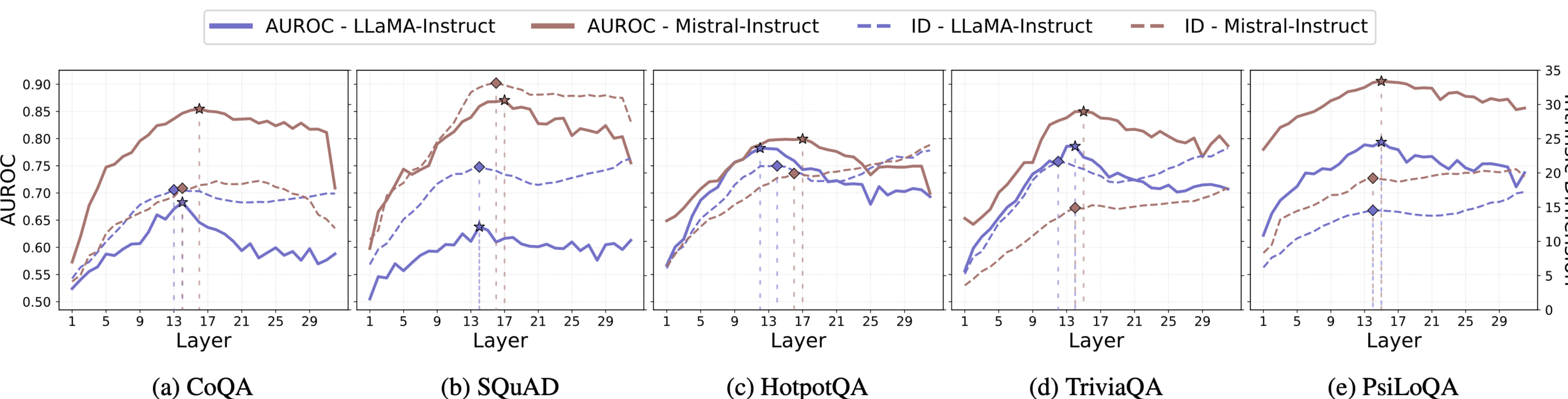
- 1st peak · abstract semantic information
- 2nd peak · lexical or surface-level information



1. Identify all local maximum of $\{d_{ID}^{(\ell)}\}_{\ell=1}^L$
2. Discard candidate peak at layer ℓ if ID monotonically increase and exceed $d_{ID}^{(\ell)}$ within a window of size w .
3. Choose the earliest remaining peak

Experimental Results of Layer Selection

LLaMA-3.1-8B-Instruct							Mistral-7B-Instruct-v0.3						
	CoQA	SQuAD	HotpotQA	TriviaQA	PsiLoQA	Avg	CoQA	SQuAD	HotpotQA	TriviaQA	PsiLoQA	Avg	
Pred. Entropy	0.5833	0.5703	0.7103	0.6859	0.3604	0.5820	0.7200	0.7316	0.6303	0.6763	0.6110	0.6738	
LN-Pred. Entropy	0.5781	0.5671	0.7087	0.6774	0.3555	0.5774	0.6528	0.6390	0.6589	0.6828	0.4418	0.6151	
Semantic Entropy	0.5003	0.5518	0.4454	0.5505	0.6076	0.5311	0.5769	0.6409	0.6418	0.7353	0.6853	0.6560	
Lexical Similarity	0.6780	0.5988	0.7294	0.6838	0.4082	0.6196	0.7071	0.7169	0.6946	0.7547	0.5781	0.6903	
LID	0.5059	0.5281	0.5171	0.4989	0.5994	0.5299	0.5274	0.5688	0.5518	0.4970	0.6447	0.5579	
EigenScore	0.5247	0.5300	0.5987	0.5882	0.5080	0.5499	0.7092	0.7508	0.6587	0.7273	0.7246	0.7141	
RankME	0.6598	0.6071	0.7040	0.7114	0.7478	0.6860	0.7743	0.7341	0.6909	0.6699	0.8277	0.7394	
Curvature	0.6323	0.6183	0.7413	0.7366	0.7565	0.6970	0.8492	0.8553	0.7940	0.8368	0.9005	0.8472	
Val Loss	0.6705	0.6164	0.7682	0.7861	0.7836	0.7250	0.8283	0.8679	0.7968	0.8496	0.8861	0.8457	
RGN	0.5993	0.6130	0.7040	0.7859	0.7373	0.6879	0.7090	0.7553	0.7493	0.7868	0.8562	0.7713	
SNR	0.5240	0.5699	0.7009	0.5567	0.6624	0.6028	0.7475	0.6870	0.7620	0.7008	0.8207	0.7436	
ID	0.6705	0.6130	0.6932	0.7073	0.7373	0.6843	0.8283	0.8679	0.6993	0.7868	0.8533	0.8071	
FEPoID	0.6705	0.6377	0.7807	0.7516	0.7862	0.7253	0.8466	0.8679	0.7982	0.8496	0.9031	0.8531	



Best detection performance · Consistently selects **optimum or near-optimum** layer

LLaMA-3.1-8B-Instruct						Mistral-7B-Instruct-v0.3						Llama-3.1-8B			Llama-3.2-3B			Llama-3.2-1B		
	HaluEval	CNN/DM	Avg	HaluEval	CNN/DM	Avg		HaluEval	CNN/DM	Avg										
RankMe	0.6075	0.5774	0.5924	0.7149	0.6869	0.7009	RankMe	0.7606	0.7087	0.6561										
Curvature	0.5494	0.5922	0.5708	0.7498	0.7319	0.7409	Curvature	0.8410	0.8113	0.7692										
Val Loss	0.5961	0.5859	0.5910	0.7294	0.6938	0.7116	Val Loss	0.8523	0.8323	0.7800										
RGN	0.5713	0.5821	0.5767	0.7563	0.7031	0.7297	RGN	0.7671	0.7187	0.7078										
SNR	0.5528	0.5474	0.5501	0.7385	0.6811	0.7098	SNR	0.6943	0.7550	0.7034										
ID	0.6075	0.5918	0.5997	0.7498	0.7185	0.7342	ID	0.8210	0.7912	0.7402										
FEPoID	0.6165	0.5995	0.6080	0.7808	0.7614	0.7711	FEPoID	0.8581	0.8339	0.7860										

Generalizes across QA and summarization tasks, as well as model scales and tuning strategies

	CoQA	SQuAD	HotpotQA	TriviaQA	PsiLoQA	Avg
RankMe	26.32	27.65	28.98	28.17	25.41	27.30
Curvature	43.88	45.23	45.55	46.16	45.37	45.24
Val Loss	27.59	30.81	29.80	29.79	29.78	29.55
RGN	54.83	60.95	58.31	58.08	58.72	58.18
SNR	54.20	59.91	57.31	58.84	59.16	57.88
FEPoID; ID	9.40	9.90	9.95	10.59	10.88	10.14

The best time efficiency

3 times faster
than Validation Loss

Q2: Which Token Should We Extract ?

Question: Were Scott Derrickson and Ed Wood of the same nationality?

LLaMA-Instruct: **No**. Scott Derrickson is American, Ed Wood is American but of English descent. However, both are American. So the answer is **yes**

Mistral-Instruct: No, Scott Derrickson is American and Ed Wood is Canadian.

(a) Inconsistent Continuation

Question: When was the American writer of the a series of four vampire-themed fantasy romance novels **born**?

LLaMA-Instruct: Anne Rice was born on October 4, 1941. The series is called "The Vampire Chronicles". **The first book in the series**

Mistral-Instruct: Cassandra Clare was born on July 27, 1971.

(b) Semantic Drift

Question: In what corner of Missouri is radio station WMBH located?

LLaMA-Instruct: The Ozark Mountains. The station is located in the Ozark Mountains of southern Missouri. **The station is located in the Ozark Mountains of**

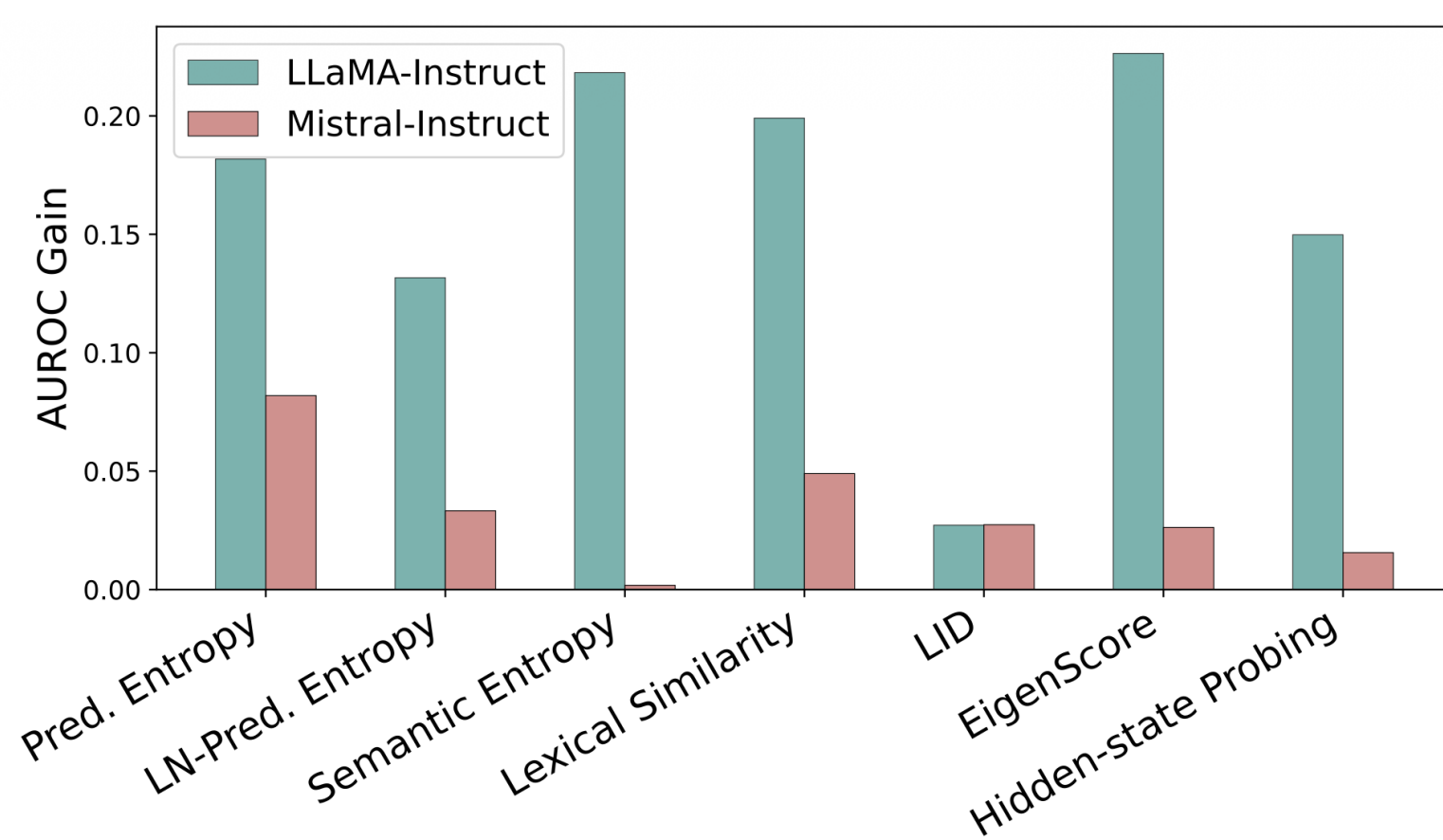
Mistral-Instruct: WMBH is located in the southeast corner of Missouri.

(c) Degenerate Repetition

End-of-sequence Noise

First-Sentence Truncation (FST)

LLaMA-Instruct: The Ozark Mountains. **The station is located in the Ozark Mountains of southern Missouri. The station is located in the Ozark Mountains of**



Improves all methods

Benefits multiple levels of information used for hallucination detection

Dataset	Fisher Separation ↑ w/o FST	Fisher Separation ↑ w/ FST	Silhouette Score ↑ w/o FST	Silhouette Score ↑ w/ FST
CoQA	0.0003	0.0044	-0.0007	0.0324
SQuAD	0.0004	0.0049	0.0006	0.0973
HotpotQA	0.0009	0.0075	-0.0001	-0.0011
TriviaQA	0.0013	0.0185	0.0059	0.1006
PsiLoQA	0.0016	0.0340	0.0118	0.2605

FST yields more **separable representations** for hallucination detection